observations:

kinematic observation

x, y → world for vehicle, (x,y) for other into vehicle

v_x, v_y → velocity of vehicle in x & y axes

ψ → heading angle

here I need the kinematic observation

action space → continuous action space → directly control the throttle & δ

or a discrete

Now: → continuous

Dynamics

$\dot{x} = v\cos(\psi + \beta)$

$\dot{y} = v\sin(\psi + \beta)$

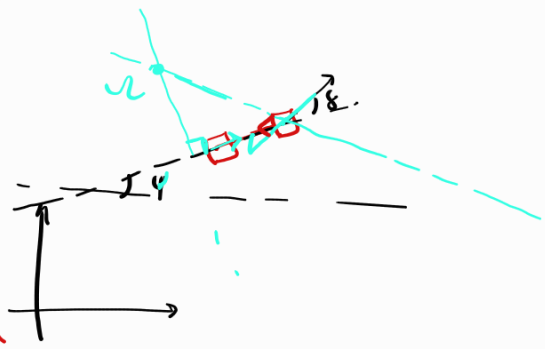
$\dot{v} = a = 0$

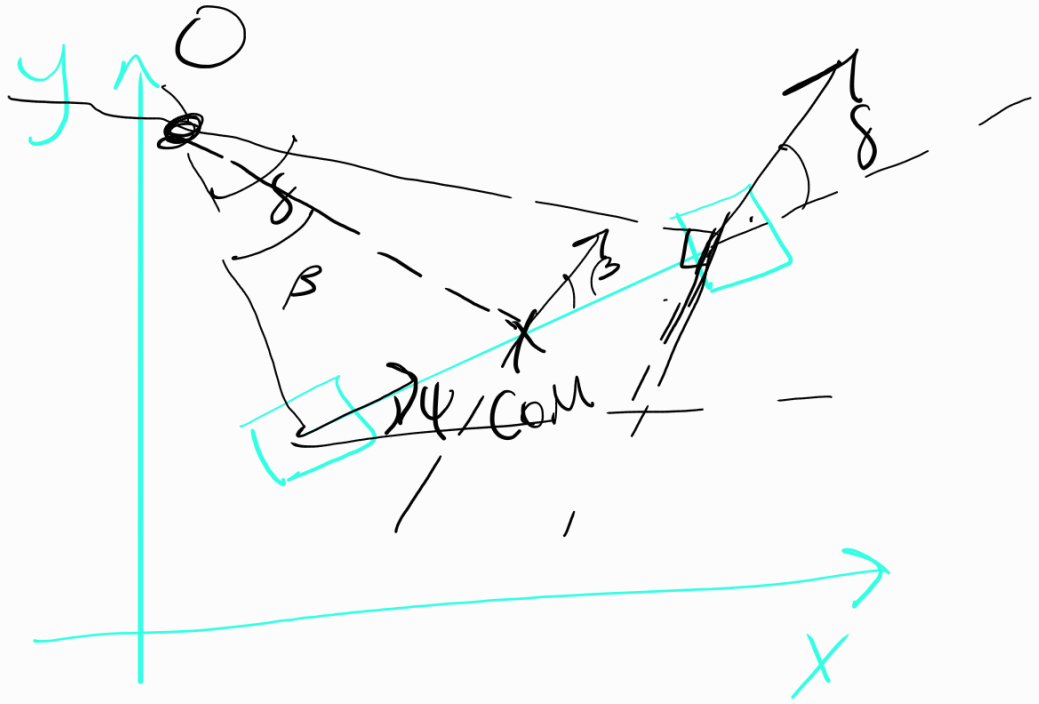
$\dot{\psi} = \frac{v}{l}\sin(\beta)$ → steering yaw

* lane: parameterized center line curve

now even with the lane: caught back

↳ world offset / vehicle offset





$\boxed{\delta : \text{to be learned}}$

$\left\{ V = \text{velocity of COM} \right\}$

$\left\{ \begin{array}{l} R : \text{radius of curvature} \\ \bar{R} : \text{radius of curvature CRA} \end{array} \right\}$

$\left\{ \beta : \text{body slip angle} \right\}$

$l_f \quad l_R \quad \text{COM}$

$\dot{\psi} = h(\delta, v)$?

ND

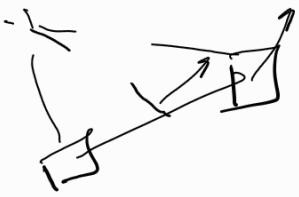
↳ smooth derivative

↳ would get a smooth curve

$$\dot{x} = \frac{dx}{dt} = V\cos(\psi + \beta)$$

$$\dot{y} = \frac{dy}{dt} = V\sin(\beta + \psi).$$

now



so $\frac{AB}{OB} = \tan(\delta) = \frac{l_f + l_r}{\overline{OB}}$

motion of any point on a rigid body can be expressed as a rotation about an ICR ....

now

then $\dot{\psi} = \frac{v}{l} \sin(\beta)$

where $\beta = \tan^{-1}\left(\frac{1}{2}\tan(\delta)\right)$ → hmmm ..... ?

slip angle

steering angle

longitudinal controller

$a = k_p (v_d - v)$

desired

input

now

position control:

$V_{lat,r} = -k_{p\,lat} \Delta_{lat}$

$\Delta\psi_r = \sin^{-1}\left(\frac{V_{lat,r}}{v}\right)$

$\psi_r = \psi_L + \Delta\psi_r$

$\dot{\psi}_r = k_{p\psi}(\psi_r - \psi)$

$\delta = \sin^{-1}\left(\frac{1}{2} l \frac{\dot{\psi}_r}{v}\right)$

$\psi_r = \psi_L + \Delta\psi_r$

$\dot{\psi}_r = k_{p,\psi}(\psi_r - \psi)$

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observation

actions

kinematic model $\longrightarrow$ 'δ'

$\dot{\psi} = \frac{v}{L}\sin\beta$

$\beta = \tan^{-1}\left(\frac{1}{2}\tan\delta\right)$

but how is it developed

all in the step fact

$\dot{\psi} = \frac{v}{L}\tan\delta$

$\psi_r \longrightarrow$ target

2 errors $\quad e = \begin{bmatrix} e_y \\ e_\psi \end{bmatrix}$

best case $\longrightarrow$ same ψ

wrap?

ψ - - - - oh yes

$-\pi, \pi$